ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Statistics Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

STATISTICS TOOLKIT · CH6 · L21

ENG. ESLAM AHMED

The Map

This lesson collapses Statistics Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Choose the average

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use frequency table totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Estimate the mean

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Interpret spread

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 58 website questions, 58 checked solutions, 4 local PDFs read.

01 Choose the average

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean $= \frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 4 The table shows information about the weights, in kilograms, of 40 babies. Weight (w kg) Frequency $2 < w < 3$ 12 $3 < w < 4$ 16 $4 < w < 5$ 9 $5 < w < 6$ 2 $6 < w < 7$ 1 (a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) One of the 40 babies is going to be chosen at random. (c) Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

02 Use frequency table totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean $= \frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 4 The table shows information about the weights, in kilograms, of 40 babies. Weight (w kg) Frequency $2 < w < 3$ 12 $3 < w < 4$ 16 $4 < w < 5$ 9 $5 < w < 6$ 2 $6 < w < 7$ 1 (a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) One of the 40 babies is going to be chosen at random. (c) Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

03 Estimate the mean

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 < w \leq 4$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q4

(a) The greatest frequency is

$$3 < w \leq 4$$

(b) Use class midpoints

$$\frac{2.5(12) + 3.5(16) + 4.5(9) + 5.5(2) + 6.5(1)}{40} = 3.6$$

Finish: (a) $3 < w \leq 4$, (b) 3.6 kg, (c) $\frac{3}{10}$.

FORMULA BANK

Mean $= \frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 4 The table shows information about the weights, in kilograms, of 40 babies. Weight (w kg) Frequency 2 < w < 3 12 3 < w < 4 16 4 < w < 5 9 5 < w < 6 2 6 < w < 7 1 (a) Write down the modal class. (1) (b) Work out an estimate for the mean weight of the 40 babies. kg (4) One of the 40 babies is going to be chosen at random. (c) Find the probability that this baby has a weight of more than 5 kg. (2)

YOUR SOLUTION

04 Interpret spread

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5,\ 7,\ 11,\ 12,\ 13,\ 14,\ 15,\ 16,\ 17,\ 18,\ 18$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q14

Put the marks in order

$$5,\ 7,\ 11,\ 12,\ 13,\ 14,\ 15,\ 16,\ 17,\ 18,\ 18$$

Finish: 6.

FORMULA BANK

Mean $= \frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 14 Max kept a record of the marks he scored in each of the 11 spelling tests he took one term. Here are his marks. 18 5 7 12 11 18 15 16 17 13 14 Find the interquartile range of the marks.

YOUR SOLUTION

05 Calculate statistic

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Statistics Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$a = 7$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q2

Calculate statistic

$$a = 7$$

The median of four numbers

$$\frac{a+b}{2}$$

Finish: $a = 7,\ b = 10,\ c = 12.$

FORMULA BANK

Mean $= \frac{\sum fx}{\sum f}$. Median is middle. Modal class has highest frequency.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 2 Given that $a < b < c$ the four whole numbers a, a, b and c have a mode of 7 a median of 8.5 a mean of 9 Work out the value of a, the value of b and the value of c. a = b = c =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Choose the average
"work out", "show", "hence", a familiar exam phrase	Use frequency table totals
"work out", "show", "hence", a familiar exam phrase	Estimate the mean
"work out", "show", "hence", a familiar exam phrase	Interpret spread
"work out", "show", "hence", a familiar exam phrase	Calculate statistic

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Histograms

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

HISTOGRAMS · CH6 · L18

ENG. ESLAM AHMED

The Map

This lesson collapses Histograms into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use frequency density

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find missing frequency

TRIGGER *"work out", "find", a missing value*

03 Read area under a bar

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Build the histogram scale

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use histogram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Use frequency table

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 33 website questions, 33 checked solutions, 3 local PDFs read.

01 Use frequency density

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5(40) + 10(20) = 400$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q19

Use the relative histogram heights

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate area

$$5(40) + 10(20) = 400$$

Finish: 370 students.

FORMULA BANK

Frequency density $= \frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

02 Find missing frequency

TRIGGER *"work out", "find", a missing value*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5(40) + 10(20) = 400$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?

2 Write the rule, theorem, or equation before any calculator work.

3 Substitute one value at a time and keep units/unknowns visible.

4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q19

Use the relative histogram heights

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate area

$$5(40) + 10(20) = 400$$

Finish: 370 students.

FORMULA BANK

Frequency density $= \frac{\text{frequency}}{\text{class width}}$. Frequency = density × width.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

03 Read area under a bar

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1,\ 4,\ 9,\ 6,\ 1$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q18

Use histogram

$$1,\ 4,\ 9,\ 6,\ 1$$

Use histogram

$$28,\ 112,\ 126,\ 84,\ 42$$

Finish: 4.36 hours.

FORMULA BANK

Frequency density $= \frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

04 Build the histogram scale

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?
2 Write the rule, theorem, or equation before any calculator work.
3 Substitute one value at a time and keep units/unknowns visible.
4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 4WM1H Q20

Frequency density

$$0 < h \leq 5 : \frac{12}{5} = 2.4$$

Finish: draw bars with densities $2.4, 3.4, 3, 1.4$.

FORMULA BANK

Frequency density $= \frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

05 Use histogram

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1,\ 4,\ 9,\ 6,\ 1$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q18

Use histogram

$$1,\ 4,\ 9,\ 6,\ 1$$

Use histogram

$$28,\ 112,\ 126,\ 84,\ 42$$

Finish: 4.36 hours.

FORMULA BANK

Frequency density $= \frac{\text{frequency}}{\text{class width}}$. Frequency = density $\times$ width.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

06 Use frequency table

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Histograms question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1HR Q21

Use histogram

$$\text{frequency} = \text{class width} \times \text{frequency density}$$

Use frequency table

$$\text{frequency density} = \frac{16}{0.5} = 32$$

Finish: 102 watermelons

FORMULA BANK

Frequency density $= \frac{\text{frequency}}{\text{class width}}$. Frequency = density × width.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> A class interval $20 < x \leq 35$ has frequency 24. Work out the frequency density, then draw the bar height.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use frequency density
"work out", "find", a missing value	Find missing frequency
"work out", "show", "hence", a familiar exam phrase	Read area under a bar
"work out", "show", "hence", a familiar exam phrase	Build the histogram scale
"work out", "show", "hence", a familiar exam phrase	Use histogram
"work out", "show", "hence", a familiar exam phrase	Use frequency table

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Cumulative Frequency Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CUMULATIVE FREQUENCY DIAGRAMS · CH6 · L17

ENG. ESLAM AHMED

The Map

This lesson collapses Cumulative Frequency Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Build cumulative totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Read median and quartiles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Find IQR

TRIGGER *"work out", "find", a missing value*

04 Use percentages from the curve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use cumulative frequency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 24 website questions, 24 checked solutions, 3 local PDFs read.

01 Build cumulative totals

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$Q_1 \text{ is at cumulative frequency } 20$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q12

There are people
This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

$$Q_1 \text{ is at cumulative frequency } 20$$

Finish: IQR about 18 minutes, and about 70 men.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 12 A total of 80 men and women took part in a race. The cumulative frequency graph gives information about the times, in minutes, they took for the race. 80 70 60 50 40 30 20 10 020 30 40 50 60 70 Time (minutes) Cumulative frequency (a) Use the graph to find an estimate for the interquartile range. minutes (2) 60% of the men took 50 minutes or less for the race. No women took 50 minutes or less for the race. (b) Work out an estimate for the number of men who took part in the race. (3)

YOUR SOLUTION

02 Read median and quartiles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{median} \approx 22.5 \text{ minutes}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jan 2020 P1HR Q12
>
> **There are people**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **The median is at cumulative**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** median about 22.5 minutes, IQR about 12 minutes. Hospital A is quicker and more consistent.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4$, $3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 12 The cumulative frequency graph gives information about the waiting times, in minutes, of people with appointments at Hospital A. 60 50 40 30 20 10 0 50 40 30 20 10 0 Waiting time (minutes) Cumulative frequency (a) Use the graph to find an estimate of the median waiting time at Hospital A. minutes (1) (b) Use the graph to find an estimate of the interquartile range of the waiting times at Hospital A. minutes (2) At a different hospital, Hospital B, the median waiting time is 28 minutes and the interquartile ...

YOUR SOLUTION

03 Find IQR

TRIGGER *"work out", "find", a missing value*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{median} \approx 146 \text{ g}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q11

Use cumulative frequency
This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

$$\text{median} \approx 146 \text{ g}$$

Finish: median about 146 g, and $d \approx 150$.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4$, $3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 11 The cumulative frequency graph gives information about the weights, in grams, of 90 bags of sweets. 120 90 80 70 60 50 40 30 20 10 0 Weight (grams) Cumulative frequency 130 140 150 160 170 180 (a) Find an estimate for the median of the weights of these bags of sweets. grams (2) Roberto sells the bags of sweets to raise money for charity. Bags with a weight greater than d grams are labelled large bags and sold for 3.75 euros each bag. The total amount of money he receives by selling all the large bags is 93.75 ...

YOUR SOLUTION

04 Use percentages from the curve

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

> Write the main rule first, then substitute the values.

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** May 2025 4WM2HR Q11
>
> **Required**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Median at the th value**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** Alison is more consistent — her IQR (15 km) is smaller than Tomas's (28 km), meaning less variability in daily distances.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4$, $3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 11 The cumulative frequency graph shows information about the distance, in kilometres, Ehrick drives each day for 60 days. 10 0 20 30 40 50 60 50 40 30 60 70 80 distance (kilometres) Cumulative frequency 90 (a) Use the graph to find an estimate for the median of the distances. kilometres (1) (b) Use the graph to find an estimate for the interquartile range of the distances. kilometres (2) (c) Work out the percentage of days that Ehrick drives for more than 75 kilometres. Give your answer correct to one decimal ...

YOUR SOLUTION

05 Use cumulative frequency

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$Q_1 \text{ is at cumulative frequency } 20$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q12

There are people

This is a reasoning step: write one clear sentence, then continue the calculation.

Use cumulative frequency

$$Q_1 \text{ is at cumulative frequency } 20$$

Finish: IQR about 18 minutes, and about 70 men.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4$, $3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 12 A total of 80 men and women took part in a race. The cumulative frequency graph gives information about the times, in minutes, they took for the race. 80 70 60 50 40 30 20 10 020 30 40 50 60 70 Time (minutes) Cumulative frequency (a) Use the graph to find an estimate for the interquartile range. minutes (2) 60% of the men took 50 minutes or less for the race. No women took 50 minutes or less for the race. (b) Work out an estimate for the number of men who took part in the race. (3)

YOUR SOLUTION

06 Convert standard form

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Cumulative Frequency Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7,\ 33,\ 57,\ 71,\ 78,\ 80$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q12

Complete the cumulative frequencies

$$7,\ 33,\ 57,\ 71,\ 78,\ 80$$

Plot the points

$$(0,0),(10,7),(20,33),(30,57),(40,71),(50,78),(60,80)$$

Finish: cumulative frequencies $7,33,57,71,78,80$, median about 23 minutes, probability about 0.095.

FORMULA BANK

Plot upper class boundary against cumulative frequency. Median at $N/2$, quartiles at $N/4, 3N/4$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 12 The table gives information about the times, in minutes, taken by 80 customers to do their shopping in a supermarket. Time taken (t minutes) Frequency 0 < t < 10 7 10 < t < 20 26 20 < t < 30 24 30 < t < 40 14 40 < t < 50 7 50 < t < 60 2 (a) Complete the cumulative frequency table. Time taken (t minutes) Cumulative frequency 0 < t < 10 0 < t < 20 0 < t < 30 0 < t < 40 0 < t < 50 0 < t < 60 (1) (b) On the grid opposite, draw a cumulative frequency graph for your table. 0 10 20 30 40 50 60 Cumulative frequency ...

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Build cumulative totals
"work out", "show", "hence", a familiar exam phrase	Read median and quartiles
"work out", "find", a missing value	Find IQR
"work out", "show", "hence", a familiar exam phrase	Use percentages from the curve
"work out", "show", "hence", a familiar exam phrase	Use cumulative frequency
"work out", "show", "hence", a familiar exam phrase	Convert standard form

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Probability Toolkit

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PROBABILITY TOOLKIT · CH6 · L20

ENG. ESLAM AHMED

The Map

This lesson collapses Probability Toolkit into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

02 Use complement

TRIGGER *"probability", "random", "given that"*

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 45 website questions, 45 checked solutions, 3 local PDFs read.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{1}{x}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1HR Q24

There are counters. Initially 5
This is a reasoning step: write one clear sentence, then continue the calculation.

List probability cases
This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $x = 8$

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

02 Use complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$r + 0.24 + 2r + 0.31 = 1$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q6

Let the probability of red

$$r + 0.24 + 2r + 0.31 = 1$$

Expected number of red counters

$$180 \times 0.15 = 27$$

Finish: 27.

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2HR Q21

There are beads altogether
This is a reasoning step: write one clear sentence, then continue the calculation.

Red beads

$$6$$

Finish: $\boxed{18}$

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{no lemon}) = \frac{11}{16} \times \frac{10}{15} \times \frac{9}{14}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q18

There are sweets and are

$$P(\text{no lemon}) = \frac{11}{16} \times \frac{10}{15} \times \frac{9}{14}$$

Finish: $\frac{33}{112}$.

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Toolkit question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$0.65 \times 300 = 195$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jan 2022 P1HR Q2
>
> **Calculate probability**
>
> $$0.65 \times 300 = 195$$
>
> **Finish:** 195.

FORMULA BANK

$P(A) = \frac{\text{wanted}}{\text{total}}$. Complement: $P(\text{not } A) = 1 - P(A)$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"probability", "random", "given that"	Count wanted over total
"probability", "random", "given that"	Use complement
"probability", "random", "given that"	Multiply along tree branches
"probability", "random", "given that"	Use conditional probability
"probability", "random", "given that"	Calculate probability

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Probability Diagrams - Venn & Tree Diagrams

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PROBABILITY DIAGRAMS - VENN & TREE DIAGRAMS · CH6 · L19

ENG. ESLAM AHMED

The Map

This lesson collapses Probability Diagrams - Venn & Tree Diagrams into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

02 Use complement

TRIGGER *"probability", "random", "given that"*

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 29 website questions, 29 checked solutions, 3 local PDFs read.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(A \text{ odd}) = \frac{7}{10}, \qquad P(A \text{ even}) = \frac{3}{10}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q16

Calculate probability

$$P(A \text{ odd}) = \frac{7}{10}, \qquad P(A \text{ even}) = \frac{3}{10}$$

Calculate probability

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{7}{15}$$

Finish: $\frac{7}{15}$, and 25.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

02 Use complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(A\ \text{odd}) = \frac{7}{10}, \qquad P(A\ \text{even}) = \frac{3}{10}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q16

Calculate probability

$$P(A\ \text{odd}) = \frac{7}{10}, \qquad P(A\ \text{even}) = \frac{3}{10}$$

Calculate probability

$$\frac{7}{10} \times \frac{5}{12} + \frac{3}{10} \times \frac{7}{12} = \frac{7}{15}$$

Finish: $\frac{7}{15}$, and 25.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$0.6 \times 0.9 = 0.54$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q14

Winning both frames

$$0.6 \times 0.9 = 0.54$$

Winning exactly one frame

$$0.6 \times 0.1 + 0.4 \times 0.25$$

Finish: 0.54, 0.16.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q14

Calculate probability

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

Finish: $\frac{21}{190}$.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Probability Diagrams - Venn & Tree Diagrams question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q14

Calculate probability

$$P(\text{two small}) = \frac{7}{20} \times \frac{6}{19}$$

Finish: $\frac{21}{190}$.

FORMULA BANK

Venn: fill intersections first. Tree: multiply along branches, add separate routes.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"probability", "random", "given that"	Count wanted over total
"probability", "random", "given that"	Use complement
"probability", "random", "given that"	Multiply along tree branches
"probability", "random", "given that"	Use conditional probability
"probability", "random", "given that"	Calculate probability

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Combined & Conditional Probability

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

COMBINED & CONDITIONAL PROBABILITY · CH6 · L16

ENG. ESLAM AHMED

The Map

This lesson collapses Combined & Conditional Probability into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

02 Use complement

TRIGGER *"probability", "random", "given that"*

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

06 Use the complement

TRIGGER *"probability", "random", "given that"*

BANK EVIDENCE 11 website questions, 11 checked solutions, 3 local PDFs read.

01 Count wanted over total

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

> **Source pattern:** Jun 2023 P2H Q20
>
> **There are 3 red counters**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **At least one red counter**
> This is a reasoning step: write one clear sentence, then continue the calculation.
>
> **Finish:** $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

02 Use complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{odd}) = \frac{1}{2}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P2HR Q14

The complement of at least

$$P(\text{odd}) = \frac{1}{2}$$

Finish: $\frac{3}{4}$.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

03 Multiply along tree branches

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q20

There are 3 red counters

This is a reasoning step: write one clear sentence, then continue the calculation.

At least one red counter

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

04 Use conditional probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{win tennis}) = 0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q16

Calculate probability

$$P(\text{win tennis}) = 0.6$$

Calculate probability

$$0.42 \div 0.6 = 0.7$$

Finish: 0.12.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

05 Calculate probability

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{win tennis}) = 0.6$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q16

Calculate probability

$$P(\text{win tennis}) = 0.6$$

Calculate probability

$$0.42 \div 0.6 = 0.7$$

Finish: 0.12.

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

06 Use the complement

TRIGGER *"probability", "random", "given that"*

BECOMES A Combined & Conditional Probability question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$P(\text{all 3 red are taken}) = \frac{3}{12} \cdot \frac{2}{11} \cdot \frac{1}{10}$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2H Q20

There are 3 red counters

This is a reasoning step: write one clear sentence, then continue the calculation.

At least one red counter

This is a reasoning step: write one clear sentence, then continue the calculation.

Finish: $\frac{219}{220}$

FORMULA BANK

Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$. Without replacement changes the denominator.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> A bag has 5 red counters and 7 blue counters. Two counters are taken without replacement. Find the probability that both are red.

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"probability", "random", "given that"	Count wanted over total
"probability", "random", "given that"	Use complement
"probability", "random", "given that"	Multiply along tree branches
"probability", "random", "given that"	Use conditional probability
"probability", "random", "given that"	Calculate probability
"probability", "random", "given that"	Use the complement

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.